

# Coding & Solution

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|----------------------|----------------------------------------------------------|
| <b>Date</b>          | 03 October 2026                                          |
| <b>Team ID</b>       | FITBUDDY-TEAM                                            |
| <b>Project Name</b>  | FitBuddy – AI Fitness Plan Generator using Gemini Models |
| <b>Maximum Marks</b> | 3 Marks                                                  |

## Core Implementation

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### 1. Workout Plan Generation

`generate_workout_gemini()` accepts the user profile and fitness goal/intensity, constructs the Gemini prompt and returns a structured seven-day plan.

### 2. Nutrition / Recovery Tip

`generate_nutrition_tip_with_flash()` generates a concise tip aligned with the selected fitness goal.

### 3. Feedback-Based Update

`update_workout_plan()` receives the original plan and user feedback, sends the combined context to Gemini and stores the revised result.

### 4. Persistence

`database.py` stores user information and plans through SQLite/SQLAlchemy. Original and updated plan information are retained for the application's workflow.

### 5. Web Routing

FastAPI routes connect form submissions to validation, AI generation, persistence and Jinja2 rendering.

## Team Contribution

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| <b>Member</b> | <b>Main Coding Contribution</b>                                      |
|---------------|----------------------------------------------------------------------|
| Dinesh N      | FastAPI routing, application integration and authentication          |
| Afraar M      | HTML/Jinja2 templates, CSS and dashboard UI                          |
| Deepak B      | Gemini generators, prompts and feedback update logic                 |
| Vinesh D      | Database/validation support, BMI handling, testing and documentation |